# Muhammad Shaheer Luqman

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AI researcher working in reinforcement learning, computer vision, and robot manipulation. Published across arXiv (2×) and IEEE, ICPC Gold Medalist, and summa cum laude graduate (3.90/4.00).

## Education

**National University of Computer and Emerging Sciences (FAST)** — *Karachi, Pakistan*

BS COMPUTER SCIENCE — *Sep 2021 – Jun 2025*

- CGPA: **3.90 / 4.00** (Ranked **9 / 440**)
- Summa Cum Laude Honors

## Experience

**Retrocausal** — *Remote (US-based)*

APPLIED RESEARCH ENGINEER — *Jun 2025 – Present*

- Research on robot manipulation and Vision-Language-Action systems, unsupervised action segmentation, and temporal long-video understanding — **two papers submitted** (see *Research & Publications*).
- Built and shipped end-to-end AI systems for industrial clients (incl. **BMW**), from model training to GPU deployment on **AWS** and **RunPod**.

Reinforcement Learning | Robotics | VLMs | Computer Vision | PyTorch

**Retrocausal** — *Remote (US-based)*

SOFTWARE ENGINEER INTERN — *Mar 2024 – Jun 2025*

- Built a video activity-recognition pipeline with **body pose estimation** and **vision-language models** for automated industrial workflow analysis.
- Developed **LLM prompt-engineering** systems for analysis of industrial process data.

Computer Vision | VLMs | LLMs | MediaPipe | Python

## Research & Publications

**Enhancing Human-Likeness in Reinforcement Learning Agents via Hierarchical Macro Action Quantization** — *Under Review*

*arXiv:2605.30928*

A human-like RL framework (HiMAQ) that encodes human demonstrations into a two-level hierarchy of macro actions for dexterous humanoid-hand control. On D4RL it improves human-likeness over the non-hierarchical baseline while matching or exceeding success rates across IQL, SAC, and RLPD. *View on arXiv*

Robot Manipulation | Hierarchical Action Quantization | Reinforcement Learning | Imitation Learning

**Unsupervised Skeleton-Based Action Segmentation via Hierarchical Spatiotemporal Vector Quantization** — *Under Review*

*arXiv:2604.15196*

A hierarchical spatiotemporal vector-quantization framework for unsupervised skeleton-based action segmentation, achieving state-of-the-art on the HuGaDB, LARa, and BABEL benchmarks. *View on arXiv*

Unsupervised Learning | Action Segmentation | Vector Quantization | Skeleton-Based Recognition

**Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior** — *ICETST 2024*

*IEEE Xplore*

An IoT surveillance system pairing hardware sensors with AI-driven anomaly detection for real-time child-safety alerts. *View on IEEE Xplore*

IoT Systems | Anomaly Detection | Surveillance | Embedded Systems

**Research Interests:** Robot Manipulation, Computer Vision, Reinforcement Learning, Vision-Language Models, IoT + AI Systems

## Certifications

**IELTS Academic** — English Proficiency Jul 2026

Overall Band: **7.5 / 9** • Reading **9.0** • Listening **8.5**

## Technical Skills

**Languages** Python, C / C++, JavaScript.

**ML Frameworks** PyTorch, Hugging Face, Scikit-learn; YOLO, MediaPipe; Stable Diffusion; BERT, Whisper.

**Software Frameworks** Django, FastAPI, Flask; Express.js, React, Next.js, Vue.

**Developer Tools** AWS, RunPod; Docker, Git, Linux.

## Projects

**PixelForge — Generative AI Marketing Platform**

*Final Year Project (GitHub)*

Full-stack generative pipeline combining **BLIP** captioning, **Gemini** text generation, and diffusion-based image and video synthesis for product-aware marketing content.

**LushWear — Business Operation Management**

*Client (GitHub)*

**Production-ready system, built, sold, and actively maintained** for a local business to run its day-to-day operations. Full-stack inventory, order, and storefront management with a **FastAPI** backend, **Electron** desktop app, **Supabase** real-time database, and two-way **Shopify** integration.

More on my *portfolio* and *GitHub*.

## Honors & Awards

- **ICPC Gold Medalist** — ICPC National Competitive Programming *(Nov 2024)*
- **Summa Cum Laude** — Graduated with a high GPA (3.90 / 4.00), National University of Computer and Emerging Sciences (FAST) *(2025)*
- **ICPC Bronze Medalist** — ICPC National Competitive Programming *(Feb 2024)*
- **Winner** — Code Sprint, PROCOM 2023 *(2023)*
- **Best Team Award** — Coder's Cup, ACM FAST
- **4x Rector's List (4.0 GPA)** — NUCES (FAST), 4 semesters
- **4x Dean's List (3.5+ GPA)** — NUCES (FAST), 4 semesters